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Adaptive Modulation and Coding for Underwater Acoustic OFDM

Lei Wan, *Student Member, IEEE*, Hao Zhou, Xiaoka Xu, *Member, IEEE*, Yi Huang, *Student Member, IEEE*, Shengli Zhou, *Fellow, IEEE*, Zhijie Shi, *Member, IEEE*, and Jun-Hong Cui, *Member, IEEE*

Abstract—Underwater acoustic channels are fast varying spatially and temporally according to environmental conditions. Adaptive modulation and coding (AMC) is appealing for underwater acoustic communications to improve the system efficiency by matching transmission parameters to channel variations. In this paper, we construct an AMC system with a finite number of transmission modes in the context of underwater orthogonal frequency-division multiplexing (OFDM). We propose the effective signal-to-noise ratio (SNR) computed after channel estimation and channel decoding as a new performance metric for mode switching, which is shown to predict the system performance more consistently than the input SNR and the pilot SNR. Real-time AMC tests have been conducted in a recent sea experiment to maximize the transmission rate with a given transmission power.

Index Terms—Adaptive modulation and coding (AMC), effective signal-to-noise ratio (SNR), energy consumption, underwater acoustic communication.

I. INTRODUCTION

SIGNIFICANT progresses have been made on underwater acoustic communications in recent years, with both single-carrier transmissions [2]–[5] and multicarrier modulation in the form of orthogonal frequency-division multiplexing (OFDM) [1], [6]–[15]. Because underwater acoustic channels are fast varying spatially and temporally [16]–[18], the adaptive modulation and coding (AMC) technique [19, ch. 8] is appealing for underwater acoustic communications to improve

the system efficiency by varying transmission parameters according to channel conditions.

The study on the application of AMC to underwater acoustic communications has been limited in comparison to the extensive investigations on receiver designs with fixed modulations. In [20], a single-carrier phase-shift keying (PSK)-based AMC system is proposed, in which both the constellation size and the turbo code rate are adjustable. The index for different working modes is an achievable information rate with independent identically distributed (i.i.d.) Gaussian inputs and post-equalization signal-to-noise ratio (SNR). In [21], a Nakagami- m -based channel model has been proposed to simulate the channel behavior of real data sets, which is then used to predict the performance of adaptive modulation based on symbol SNR. Recently, an adaptive OFDM system that maximizes throughput under the constraint of certain target bit error rate (BER) has been proposed in [22], where the system uses the predicted channel to decide the optimal modulation sizes and power levels for different OFDM subcarriers. Sea-trial results have also been presented, with feedback implemented from a radio-frequency (RF) link [22].

In this paper, we study adaptive modulation and coding for underwater OFDM based on a finite number of transmission modes. The objective is to maximize the data rate via mode switching.

- We construct five discrete modulation and coding pairs, based on which we implement five transmission modes to an OFDM modem platform [23].
- We propose the effective SNR (ESNR) as the working mode indicator for the AMC-OFDM system. Different from input SNRs (ISNRs) and pilot SNRs (PSNRs) currently used, ESNR is calculated after channel estimation and hence it accounts for the channel estimation error. Via simulations and experimental data, we show that ESNR is a more consistent indicator of the system decoding performance than ISNRs and PSNRs under various channel conditions.
- We have carried out real-time AMC tests in a recent sea experiment near Kaohsiung, Taiwan, May 2013.

This line of research is carried out similar to the adaptive modulation and coding work in wireless radio channels, e.g., [28] in the context of wireless local area networks, but dealing with underwater acoustic channels with different characteristics.

This paper is organized as follows. Section II introduces five transmission modes and describes the incorporation of

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L. Wan, X. Xu, Y. Huang, S. Zhou are with the Department of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: lew09006@engr.uconn.edu; xuxiaoka@gmail.com; yiewhuong@gmail.com; shengli@engr.uconn.edu).

H. Zhou was with the Department of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269 USA. He is now with Aquatic Sensor Network Technology, Storrs, CT 06269 USA (e-mail: zhouhao4093@gmail.com).

Z. Shi and J.-H. Cui are with the Department of Computer Science and Engineering, University of Connecticut, Storrs, CT 06269 USA (e-mail: zshi@engr.uconn.edu; jcui@engr.uconn.edu).

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TABLE I
MODULATION AND CODING PAIRS FOR AMC-OFDM

Index	Parity check matrix size	Code rate r_c	Modulation
1	168×336	1/2	BPSK
2	336×672	1/2	QPSK
3	168×672	3/4	QPSK
4	672×1344	1/2	16QAM
5	336×1344	3/4	16QAM

those transmission modes into the existing OFDM system. Section III presents the performance metrics and the validation based on simulation and experimental data. Section IV reports the sea-trial results of the constructed AMC-OFDM system. Section V concludes this paper.

II. CONSTRUCTION OF TRANSMISSION MODES

The first task of the AMC-OFDM design is to prepare a set of transmission modes for underwater OFDM with different data rates and performance.

A. Modulation and Coding Pairs

Nonbinary low-density parity-check (LDPC) coded modulation has been adopted in [29] for underwater OFDM, where the size of the Galois field matches the constellation size. A set of modulation and coding pairs have been provided, out of which the rate 1/2, length 672 LDPC code in $GF(4)$ in combination with QPSK modulation has been implemented in an OFDM modem prototype [30].

Due to the implementation complexity of high-order Galois field, we pursue LDPC coding in $GF(4)$, and suitably match the coded symbols to binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), and 16 quadrature amplitude modulation (16-QAM) constellations. Table I lists the five transmission modes, where r_c denotes the code rate. While modes 1 and 2 are identical to those used in [29], modes 3, 4, and 5 are newly constructed. For mode 1, every coded symbol in $GF(4)$ is mapped to two BPSK symbols. For modes 2 and 3, every coded symbol in $GF(4)$ is mapped to one QPSK symbol. For modes 4 and 5, every two coded symbols in $GF(4)$ are mapped to one 16-QAM symbol, one to the real part and the other to the imaginary part. Hence, all five modulation and coding pairs in Table I lead to 672 coded symbols after coding and modulation.

Fig. 1 shows the block error rate (BLER) performance for the five modes in Table I, with both additive white Gaussian noise (AWGN) channel and i.i.d. Rayleigh fading channel with four receive elements for diversity combining. Solid lines with circle markers correspond to the simulated performance of the five modes listed in Table I, with perfect channel knowledge. About 3-dB difference can be observed between consecutive modes in the i.i.d. channel with four receivers. Dashed bold lines correspond to the capacity limit in the AWGN channel and the outage probability in the Rayleigh fading channel when the source symbols can be assumed to be Gaussian distributed. The solid bold lines correspond to the capacity limit in the AWGN channel and the outage probability in the Rayleigh fading channel when the source symbols are drawn from the same constellations as the transmission modes in Table I; see, e.g., [31]–[33] on how to compute the information-theoretic limits with a finite-alphabet

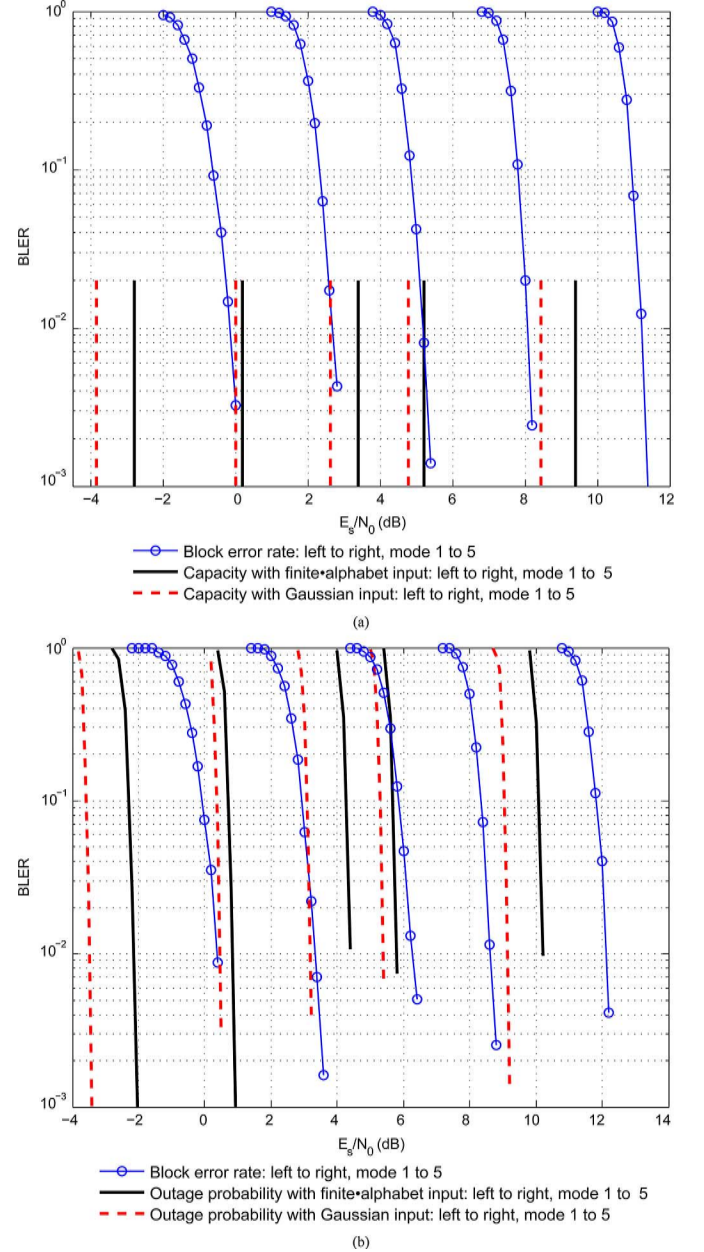


Fig. 1. BLER performance of the five modulation and coding pairs: (a) AWGN; and (b) i.i.d. Rayleigh fading.

input. With finite block lengths, the transmission modes constructed in Table I are within 2–3 dB away from the information-theoretic limits that assume infinite code length and optimal decoding. This demonstrates that the designed modulation and coding pairs have satisfactory performance.

B. Transmission Modes on an OFDM Modem Platform

Now we incorporate the modulation and coding pairs in Table I to the zero-padded OFDM (ZP-OFDM) transmission, as implemented in [30]. The OFDM bandwidth is $B = 6000$ Hz and the total number of subcarriers is $K = 1024$, which leads to a symbol duration $T_s = 170.7$ ms. Out of $K = 1024$ subcarriers, there are $K_p = 256$ pilot subcarriers reserved for channel estimation and $K_n = 96$ null subcarriers to facilitate Doppler compensation at the receiver [7]. The number of data subcarriers is $K_d = 672$, and hence each codeword can be

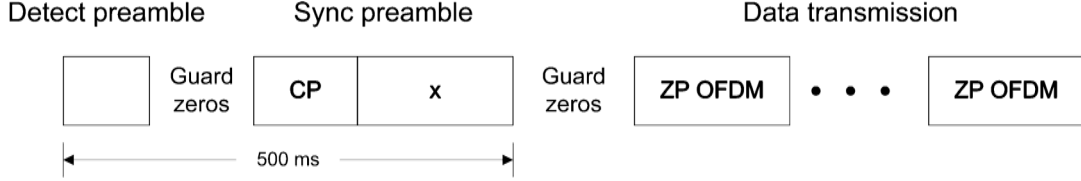


Fig. 2. Data transmission format.

TABLE II

PAYLOAD OF THE FIVE MODES WITH $T_g = 50$ ms (NOTE THAT 4 B RESERVED BY THE MODEM PHYSICAL LAYER ARE EXCLUDED IN THE COMPUTATION)

	Payload per block (bytes)	R (kb/s)
Mode 1	38	1.38
Mode 2	80	2.90
Mode 3	122	4.42
Mode 4	164	5.94
Mode 5	248	8.99

accommodated in one OFDM symbol, no matter which mode is used.

We use M as the modulation size, and T_g as the guard zero length in ZP-OFDM. The data rate R for all the modes can be calculated as

$$R = \frac{r_c K_d \log_2 M}{T_s + T_g}. \quad (1)$$

The guard interval T_g can be varied easily by the OFDM transmitter. With $T_g = 50$ ms, the corresponding data rates for the five transmission modes are listed in Table II.

We implement these five transmission modes on the OFDM modem hardware platform [23], where the transceiver algorithms run on the TMS320C6747 DSP core from Texas Instruments. As shown in Fig. 2, the OFDM signal burst used in this modem consists of one detection preamble, one synchronization preamble, and several ZP-OFDM data transmission blocks. The purpose for the detection preamble is to trigger the receiver for data processing. The functions of the synchronization preamble in the form of one cyclic-prefixed (CP) OFDM block include synchronization and conveying a limited amount of control information, which includes the transmission mode and the length of guard zeros for the ZP-OFDM data blocks to be followed. This facilitates the AMC operation as the receiver uses the correct mode parameters to decode the ZP-OFDM data blocks automatically when the transmitter switches to a different transmission mode.

C. AMC Procedure

Fig. 3 shows the AMC procedure to be tested in this paper. First, a transmitter (TX) node will initiate a request-to-send (RTS) message, and the receiver (RX) node will reply with a clear-to-send (CTS) message after receiving RTS. The CTS will report the channel conditions, upon which the TX node will choose a suitable transmission mode and transmit a data burst to the RX node.

III. PERFORMANCE METRICS

One critical component of an AMC system is to find an appropriate performance metric, based on which the transmitter can switch to a suitable transmission mode.

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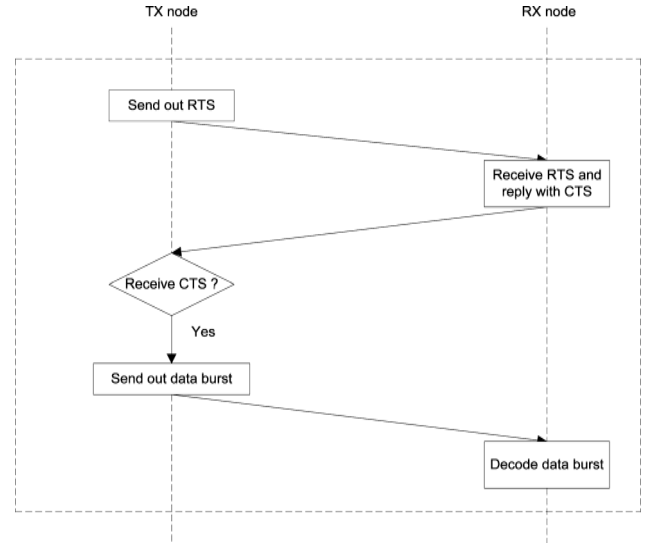


Fig. 3. AMC procedure.

A. Existing Performance Metrics

ISNR is calculated in the time domain by comparing the received signal strength to the noise strength. After synchronization, the signal and noise portions can be distinguished. Suppose that the receiver has N_r hydrophones. Denote $y_\nu[n]$ as the received sample at the signal portion and $w_\nu[n]$ as the received sample without signal transmission at the ν th hydrophone. ISNR for the ν th hydrophone is calculated as

$$\text{ISNR}_\nu = \frac{E\{|y_\nu[n]|^2\} - E\{|w_\nu[n]|^2\}}{E\{|w_\nu[n]|^2\}}. \quad (2)$$

ISNR is proportional to the received signal power. However, ISNR may not reflect the performance of OFDM in underwater channels well since it does not capture the possible intercarrier interference (ICI) due to Doppler effects.

PSNR defined in the frequency domain has been used as a measure for an underwater OFDM system in [30]. At the receiver side, after fast Fourier transform (FFT), the energy at the pilot subcarriers can serve as the indicator of useful signal strength, while the energy at null subcarriers will represent the noise strength, where the ICI is included as part of noise. Let $z_\nu[k]$ be the frequency observation at subcarrier k on the ν th hydrophone, let $\mathcal{S}_P$ be the set of pilot subcarriers, and let $\mathcal{S}_N$ be the set of null subcarriers. PSNR at the ν th hydrophone is

$$\text{PSNR}_\nu = \frac{E_{k \in \mathcal{S}_P}\{|z_\nu[k]|^2\} - E_{k \in \mathcal{S}_N}\{|z_\nu[k]|^2\}}{E_{k \in \mathcal{S}_N}\{|z_\nu[k]|^2\}}. \quad (3)$$

PSNR captures the ICI effect, however, it is computed before the channel estimation module. Given the same amount of pilot subcarriers, channels with a small number of paths could be estimated more accurately than those with a large number of paths.

Hence, PSNR might not be a consistent performance metric as it ignores the impact of channel estimation to the system performance.

B. Proposed Performance Metric

In this work, we propose to use ESNR as the performance metric. Assume there are N_r hydrophones, and define $z_\nu[k]$ to be the received frequency domain signal on the k th subcarrier on the ν th hydrophone. Correspondingly, denote $\hat{H}_\nu[k]$ as the estimated channel response of the k th subcarrier on the ν th hydrophone. Further, define the transmitted symbols on subcarrier k to be $s[k]$. ESNR is computed after the receiver has successfully decoded a message, which is used to probe the channel condition and initiate the AMC operation. During the decoding, the following channel input–output relationship has been used:

$$z_\nu[k] = \hat{H}_\nu[k]s[k] + \eta_\nu[k] \quad (4)$$

where $\eta_\nu[k]$ contains not only the ambient noise, the residual ICI, but also the noise due to the channel estimation error. The receiver uses maximum ratio combining (MRC) as follows:

$$z_{\text{mrc}}[k] = \frac{\sum_{\nu=1}^{N_r} \hat{H}_\nu^*[k]z_\nu[k]}{\sqrt{\sum_{\nu=1}^{N_r} \hat{H}_\nu^*[k]\hat{H}_\nu[k]}}. \quad (5)$$

We define $\hat{H}_{\text{mrc}}[k]$ as the equivalent channel gain

$$\hat{H}_{\text{mrc}}[k] = \sqrt{\sum_{\nu=1}^{N_r} \hat{H}_\nu^*[k]\hat{H}_\nu[k]}. \quad (6)$$

The equivalent channel input–output relationship is

$$z_{\text{mrc}}[k] = \hat{H}_{\text{mrc}}[k]s[k] + \eta_{\text{mrc}}[k]. \quad (7)$$

Correspondingly, ESNR can be calculated as

$$\text{ESNR} = \frac{E_{k \in \mathcal{S}_D} [|\hat{H}_{\text{mrc}}[k]s[k]|^2]}{E_{k \in \mathcal{S}_D} [|z_{\text{mrc}}[k] - \hat{H}_{\text{mrc}}[k]s[k]|^2]}. \quad (8)$$

Note that the average is taken over the set of data subcarriers when calculating ESNR, and $s[k]$ is known since it is done after successful decoding of the probe message.

C. Simulation Results

First, we simulate the system performance as a function of ISNR, PSNR and ESNR, respectively. The multipath channels consist of paths having the same or different Doppler scale factors

$$h(t, \tau) = \sum_{p=1}^{N_{pa}} A_p \delta(t - [\tau_p - \alpha_p t]) \quad (9)$$

where N_{pa} is the number of propagation paths, A_p is the complex amplitude of each path, τ_p is the initial delay for each path, and α_p is the Doppler scale factor for each path. The interarrival time of paths follows an exponential distribution with a mean value of 1 ms; this means that for two consecutive paths p and $p+1$, the difference between the delays τ_p and τ_{p+1} follows an exponential distribution with mean value $E[\tau_{p+1} - \tau_p] = 1$ ms. Three channel settings are used:

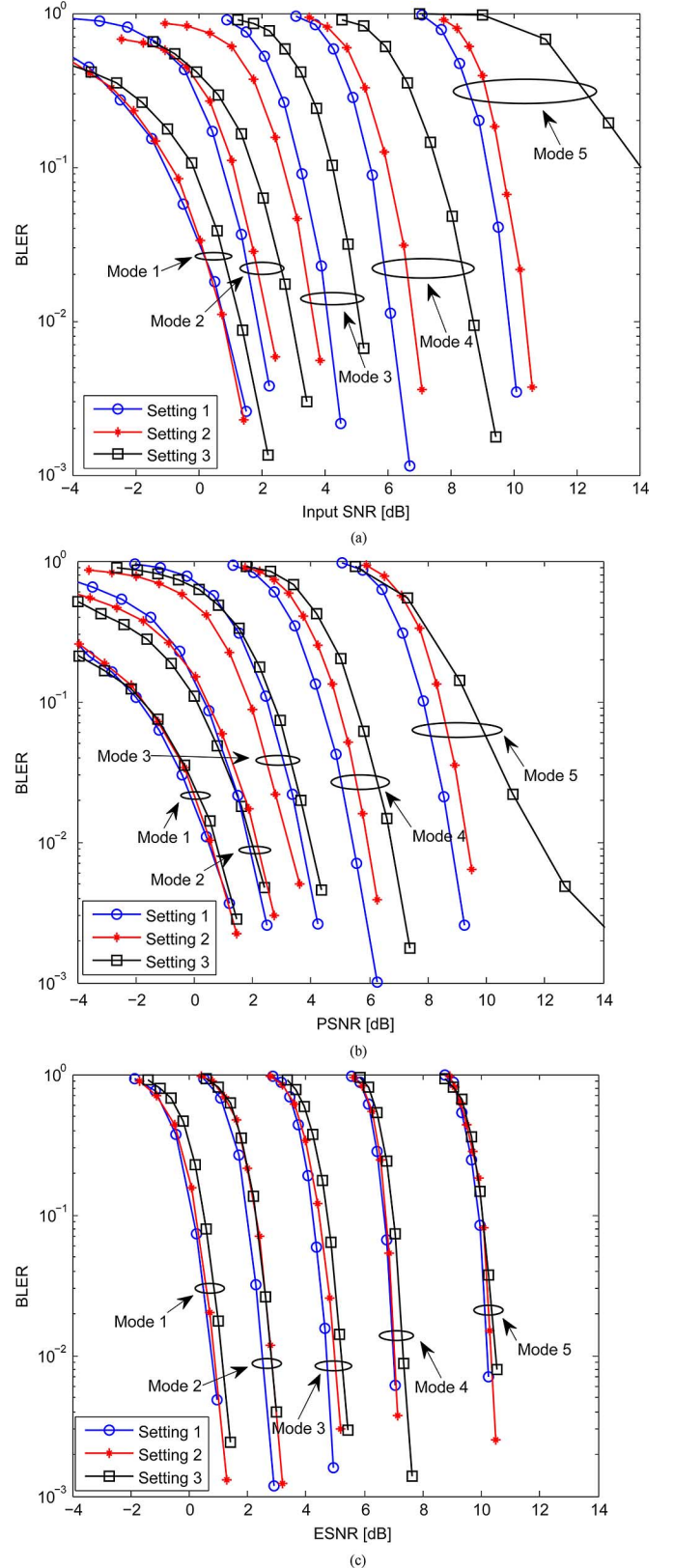


Fig. 4. BLER for five transmission modes. (a) ISNR. (b) PSNR. (c) ESNR.

- setting 1: the channel has ten paths, and the Doppler scales are zero;
- setting 2: the channel has 20 paths, and the Doppler scales are zero;

- setting 3: the channel has 20 paths, and the Doppler scales are drawn randomly from a uniform distribution $\alpha_p \sim \mathcal{U}(-10^{-4}, 10^{-4})$. The time-varying channel introduces ICI.

All the decoding performances are based on $N_r = 4$ hydrophones. ISNR and PSNR are calculated based on one hydrophone, while ESNR is calculated after combining signals from four hydrophones. Fig. 4 shows the block error rate as a function of different SNRs. The performance curves are inconsistent among three settings when ISNRs and PSNRs are used. On the other hand, the performance curves are close to each other in different channel conditions when ESNR is used. This suggests that ESNR is a more robust performance metric for AMC mode switching.

D. Performance Results With Recorded Data

In addition to simulation results, performance results based on experimental data sets are presented here to further verify that ESNR is a consistent performance metric in different channel conditions. Three experimental settings are as follows.

- Swimming pool: These data sets are collected at a swimming pool at the University of Connecticut, Storrs, CT, USA. The pool has a standard size of 25×50 m². Fig. 5(a) presents one estimated channel scattering function, which shows 2-D view of the channel (time and frequency). From Fig. 5(a), we can observe that channel length is about 20 ms. There are dense multipath arrivals due to the bounces from different sides of the pool.
- Sea test (3- and 5-km distances): On December 15, 2012, an experiment was carried out in the Long Island Sound, USA. Two sets of data corresponding to 3- and 5-km distances were collected. Fig. 5(b) shows one estimated channel scattering function corresponding to the 3-km distance, while Fig. 5(c) depicts one estimated channel scattering function corresponding to the 5-km distance. In both cases, the channels contain only one cluster.

Gaussian noise has been artificially added to the recorded data sets in order to generate BLER points corresponding to different SNRs, as shown in Fig. 6. Note that in the 3-km sea test, the mode 5 transmission was not carried out, while in the 5-km sea test, the mode 1 transmission was not carried out. Fig. 6(c) shows that the BLER-versus-ESNR performance curves obtained from different channel environments are close to each other (less than 1-dB difference) when the same transmission mode is used. This is not the case in Fig. 6(a) for the BLER-versus-ISNR curves. The BLER-versus-PSNR curves in Fig. 6(b) are also consistent in these experiments, probably because the channel conditions are relatively stable with little Doppler effect, as shown in Fig. 5. However, inconsistency has been shown in Fig. 4(b) in simulation results.

We stress that the objective in this section is not to explore how the BLER curves would shift under different SNR definitions, but rather to figure out which SNR can predict the BLER performance consistently in different channel conditions. Due to the consistency of the BLER-versus-ESNR curves, as shown here, we adopt ESNR as the performance indicator for the AMC-OFDM field test in Section IV.

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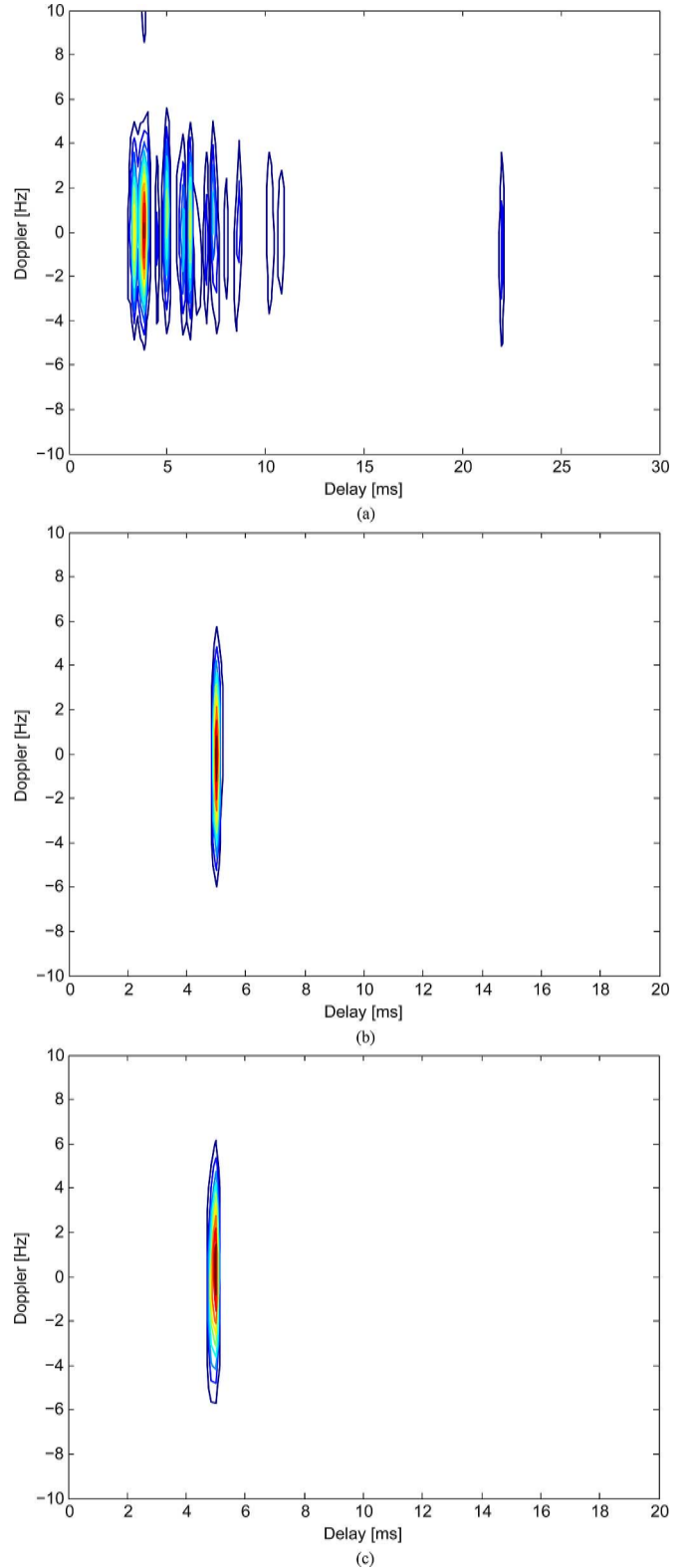


Fig. 5. Estimated channel scattering function in three different locations: (a) swimming pool; (b) sea test (3 km); and (c) sea test (5 km).

IV. REAL-TIME AMC TEST

A sea experiment was carried out in the South China Sea, near Kaohsiung City, Taiwan, from May 22 to 28, 2013. Three nodes were involved in the AMC tests on May 22, 2013, as

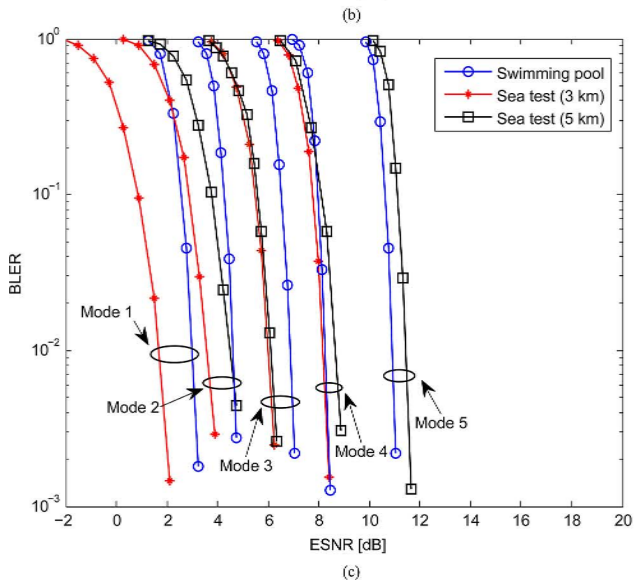
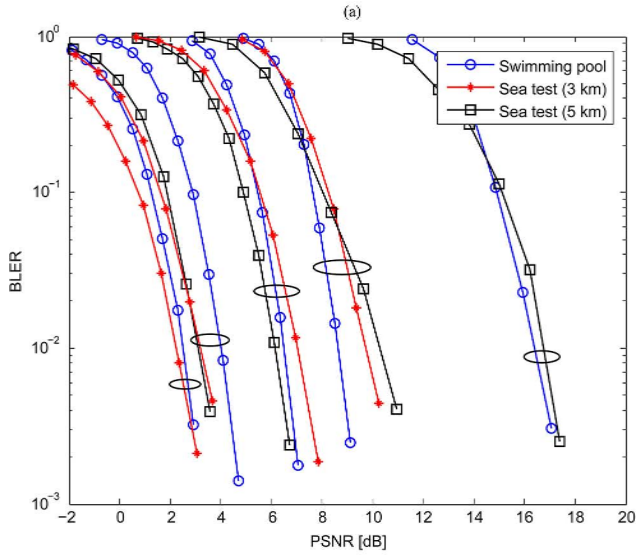
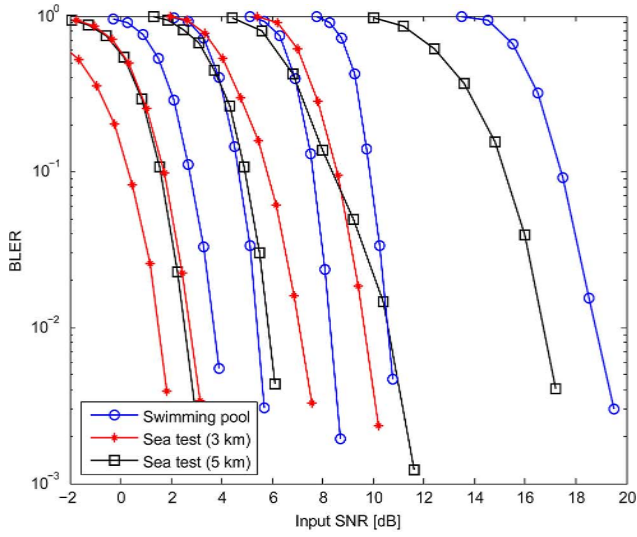


Fig. 6. Performance results using experimental data sets. (a) ISNR. (b) PSNR. (c) ESNR.

shown in Fig. 7. The distance between node 5 and node 7 was about 1038 m, and the distance between node 5 and node 9 was about 2188 m. The water depth was about 26 m, and the

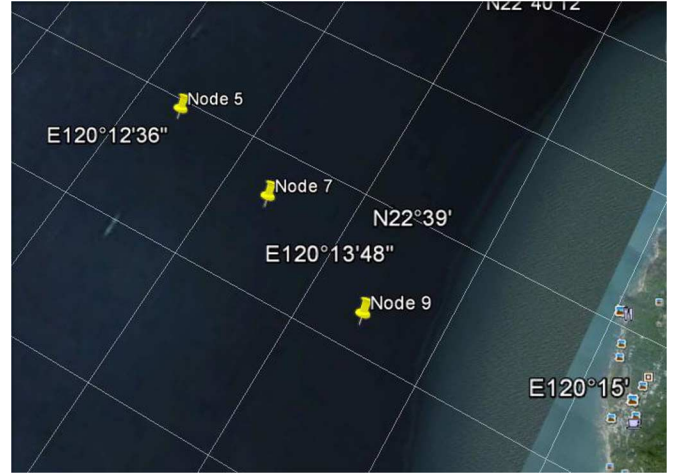


Fig. 7. Locations of the nodes used in the AMC tests.

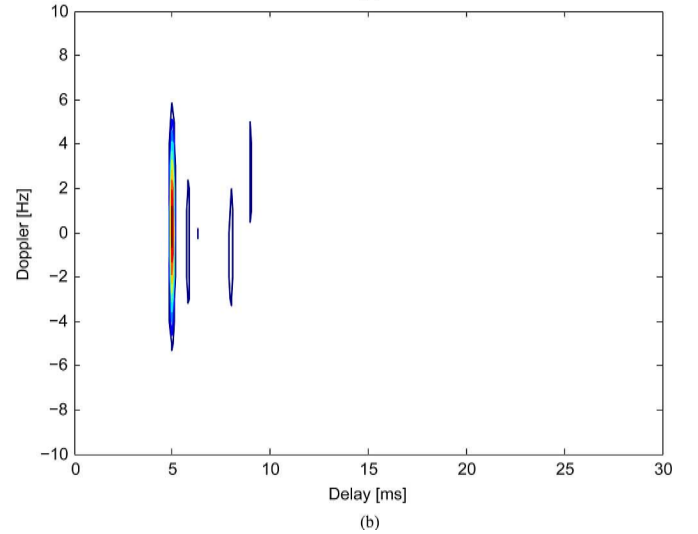
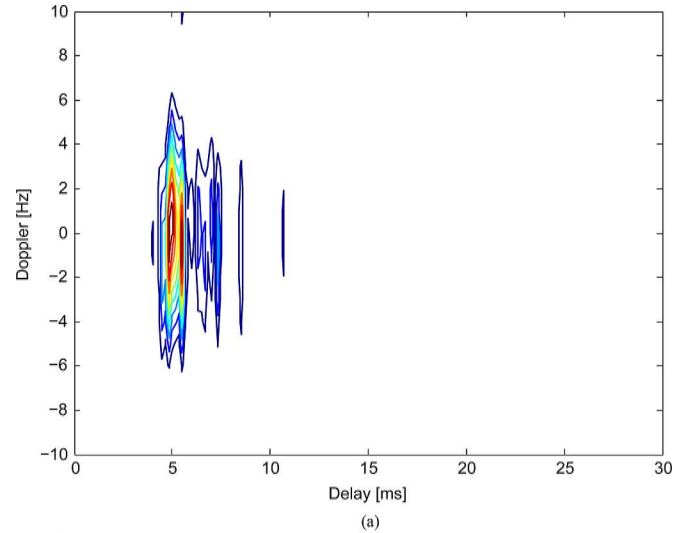


Fig. 8. Estimated channel scattering functions in the AMC tests. (a) From nodes 9 to 5. (b) From nodes 7 to 5.

modems were about 6 m below the surface. Fig. 8 shows the estimated channel scattering functions, where we can observe that the channel lengths were around 10 ms, with several multipath arrivals. In the channel from node 7 to node 5, some late arrivals have nonnegligible Doppler shifts.

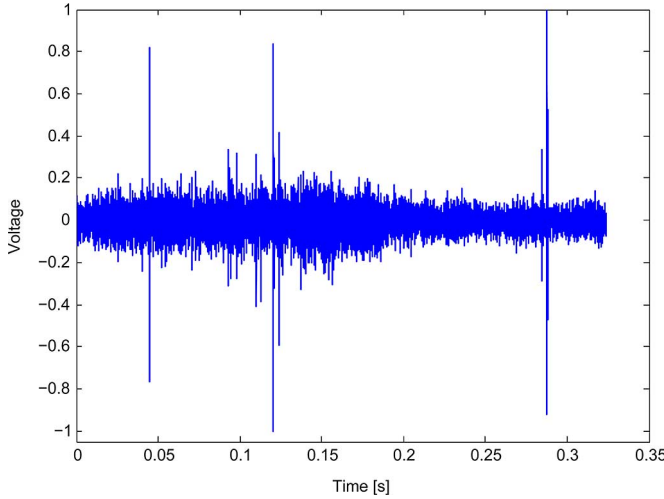


Fig. 9. Waveform affected by impulse noise.

The test procedure follows that plotted in Fig. 3. In the experiment, each data burst consisted of five OFDM data blocks. The power levels of 50%, 25%, 12.5%, 6.2%, 3.1%, and 1.5% were cycled through, so the power interval between two consecutive power levels was around 3 dB. We set the ESNR thresholds for the five transmission modes to be: 3.8, 5.0, 7.4, 9.2 and 12.2 dB. They are set according to the BLER-versus-ESNR curves in Fig. 4(c), with the target BLER 10^{-2} , plus a 2-dB protection margin.

During the experiment, there were unexpected impulsive noises, and almost all the data sets were affected with different degrees of severity. Fig. 9 shows one typical received waveform that was affected by impulse noise. This is the major reason that leads to limited values in ESNR and no transmission of modes 4 and 5 even with increased transmission power.

A. Test 1: AMC Between Nodes 9 and 5

In this test, node 5 worked as the RX node, while node 9 worked as the TX node. Table III summarizes the test performance corresponding to different transmission power levels. In Table III, R_e is the equivalent data rate, calculated as

$$R_e = \frac{\sum_{i=1}^5 N_{bl,i} N_{bi,i}}{N_{tr} T_{bl}} \quad (10)$$

where $N_{bl,i}$ is the successfully decoded data blocks in mode i ; $N_{bi,i}$ is the bits per block in mode i , calculated in Table II; N_{tr} is the total number of transmitted data blocks; and T_{bl} is one OFDM block duration (including guard zero length). Here, in this test, guard zero length is set to 150 ms, so $T_{bl} \simeq 320$ ms.

Fig. 10 shows the SNR performance according to different transmission power levels. Note that for Fig. 10(a) and (b), each circle point corresponds to one calculated SNR (ISNR or PSNR) value in one OFDM block. For Fig. 10(c), each star point corresponds to the ESNR value for one RTS, and each circle point corresponds to the ESNR value for the first OFDM block in the data burst. Because there are five data blocks in one data burst, there are much fewer circle or star points in Fig. 10(c) than in Fig. 10(a) and (b). Also in Fig. 10(c), ESNR thresholds used to distinguish different working modes have been plotted, which helps to show the distribution of AMC working modes.

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TABLE III
AMC TEST RESULTS (BETWEEN NODES 9 AND 5)

Power level		50%	25%	12.5%	6.2%	3.1%	1.5%
Mode 1	success	10	5	0	10	15	10
	failure	0	0	0	0	0	0
Mode 2	success	42	62	51	4	17	5
	failure	3	8	4	1	3	0
Mode 3	success	1	1	0	2	0	0
	failure	9	4	0	3	0	0
R_e (kb/s)		1.48	1.64	1.85	1.18	1.38	1.30
Total number of blocks		65	80	55	20	35	15

B. Test 2: AMC Between Nodes 7 and 5

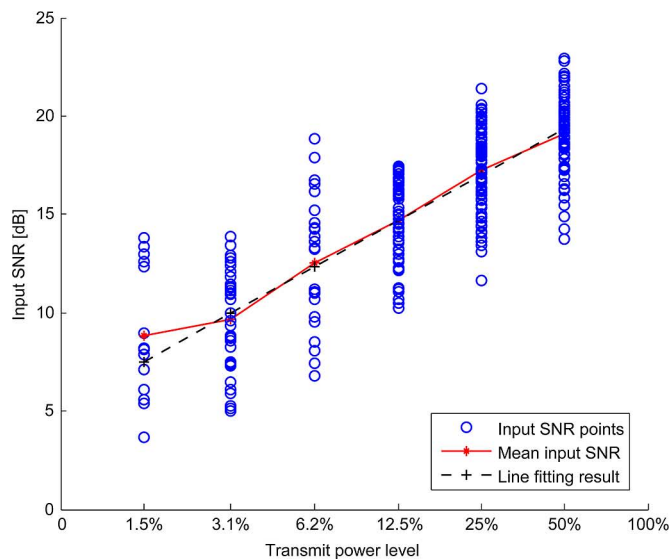
This test used node 5 as the RX node and node 7 as the TX node. Table IV presents the test results corresponding to different transmit power levels. Fig. 11 shows the SNR performance in this test. ISNRs are generally higher in test 2 than those in test 1 because the distance between RX and TX is shorter in test 2 than in test 1. PSNR and ESNR are relatively stable and remain at a low level in test 2, which explains the relative lower equivalent data rate in test 2 as compared to test 1.

From the performance results of tests 1 and 2, we have the following observations. As the transmission power increases, ISNRs and PSNRs increase. However, ESNR does not increase much as the transmission power increases, due to the existence of impulse noise, and the limitation of nonperfect channel estimation. This is why transmission modes in both tests kept almost stable no matter how transmit power varied. Although this scenario with impulse noises was unexpected before the experiment, the AMC procedure has been successfully carried out, which also shows that ESNR is an effective performance indicator for AMC mode switching.

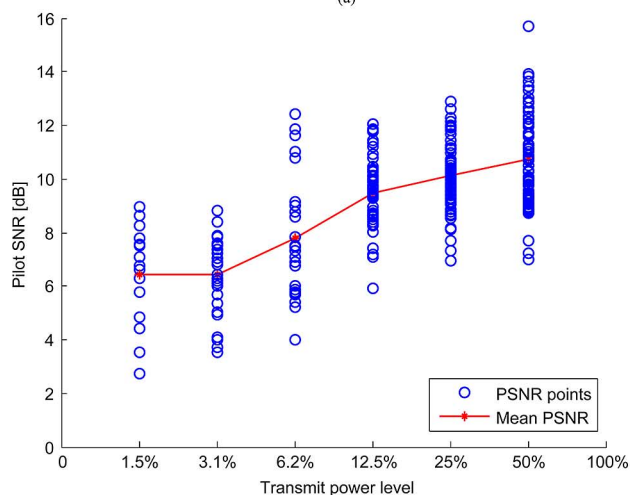
V. CONCLUSION AND DISCUSSION

This paper presented an OFDM-based AMC system for underwater acoustic communications based on a finite number of transmission modes. ESNR was proposed as a consistent performance metric for mode switching in different channel conditions. Simulation and experimental results validated the advantage of ESNR over other SNR definitions, and performance results of real-time AMC operations in a sea environment were presented.

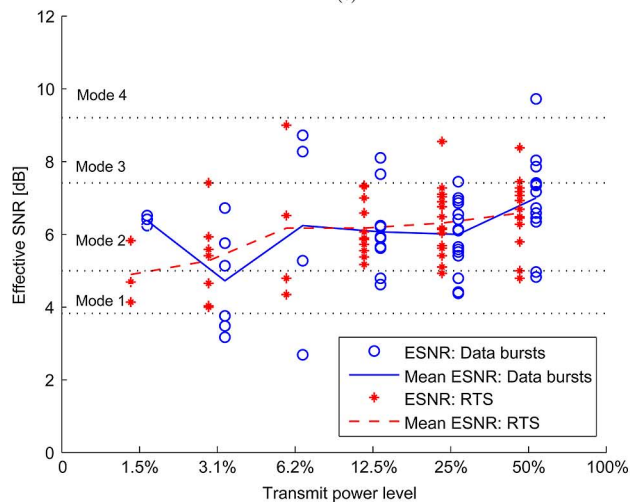
The main body of this paper focused on an AMC system with the objective to maximize the data rate. A related problem is to design an AMC system aiming to minimize the energy consumption used for transmitting a data packet with a finite number of bytes. The so-called “green communications” concept has been popular with radio communications [24]–[27]. This is an important concept for underwater systems as most of the underwater acoustic communication modems use batteries as the power source; maintaining communication while using minimum energy consumption will prolong the operation time. The challenge of energy-optimized AMC lies in how to determine the relationship between the transmission power and the receiver performance. Note that only ISNR is related to the



(a)

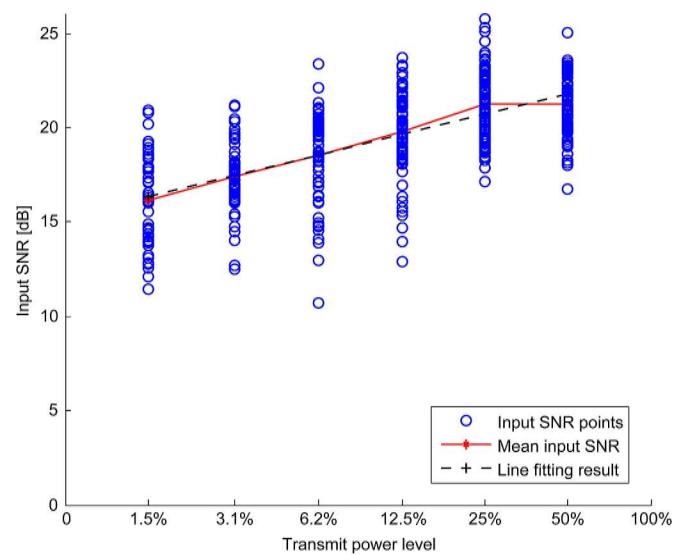


(b)

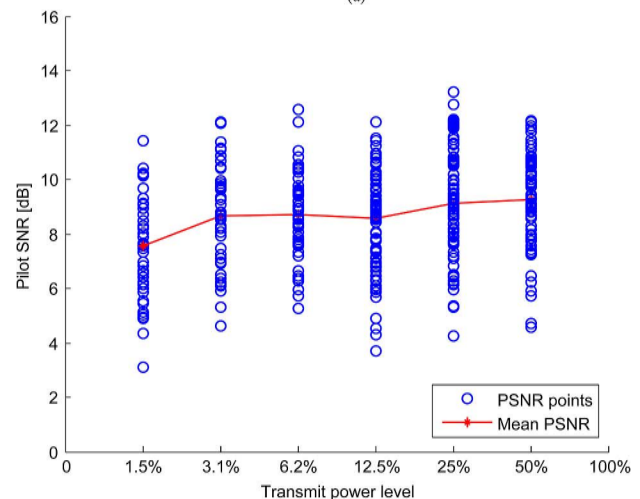


(c)

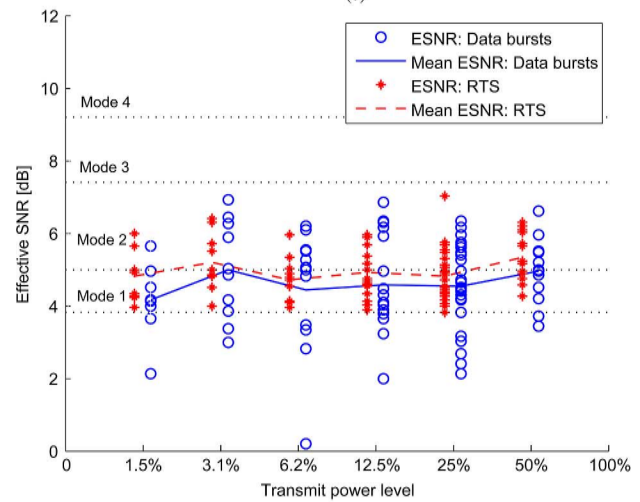
Fig. 10. SNR performance for AMC test 1. (a) ISNR of data blocks. (b) PSNR of data blocks. (c) ESNR of RTS and data.



(a)



(b)



(c)

Fig. 11. SNR performance for AMC test 2. (a) ISNR of data blocks. (b) PSNR of data blocks. (c) ESNR of RTS and data.

transmission power directly. However, the system performance does not depend on ISNRs alone, as environment conditions affect the system performance considerably. Hence, for a particular system setup in a given environment, the relationship be-

tween the BLER and ISNR needs to be established before on-line energy-optimized AMC operations. This is in contrary to ESNR which provides a consistent performance metric for rate optimization.

TABLE IV
AMC TEST RESULT (BETWEEN NODES 7 AND 5)

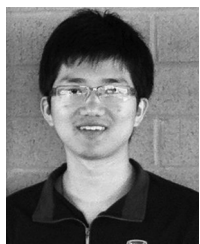
Power level		50%	25%	12.5%	6.2%	3.1%	1.5%
Mode 1	success	29	74	49	41	28	32
	failure	1	6	1	4	2	3
Mode 2	success	36	37	27	10	19	9
	failure	4	3	3	5	1	1
R_e (kb/s)		1.42	1.20	1.25	0.98	1.29	1.07
Total number of blocks		70	120	80	60	50	45

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Lei Wan (S'10) received the B.S. degree in electrical information engineering from Tianjin University, Tianjin, China, in 2006 and the M.Sc. degree in signal and information processing from Beijing University of Posts and Telecommunications (BUPT), Beijing, China, in 2009. He is currently working toward the Ph.D. degree in the Department of Electrical and Computer Engineering, University of Connecticut (UConn), Storrs, CT, USA.

His research interests include physical layer algorithm design and implementation for underwater acoustic OFDM systems.



Hao Zhou received the B.S. and Ph.D. degrees in electronic engineering from the University of Science and Technology of China (USTC), Hefei, China, in 2004 and 2009, respectively.

He is currently a Postdoctoral Researcher in the Department of Electrical and Computer Engineering, University of Connecticut (UConn), Storrs, USA. His main research interests lie in the areas of communication systems and digital signal processing. His current focus is on the implementation of orthogonal frequency-division multiplexing (OFDM)

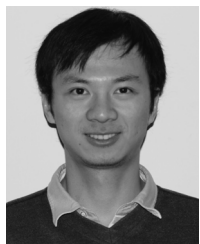
underwater acoustic modem.



Xiaoka Xu (M'13) received the B.S. degree from Henan Normal University, Xinxiang, China, in 2004 and the M.Sc. and Ph.D. degrees in signal and information processing from Harbin Engineering University, Harbin, China, in 2007 and 2009, respectively.

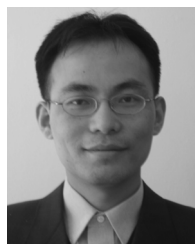
Currently he is an Assistant Research Professor with the Department of Electrical and Computer Engineering, University of Connecticut (UConn), Storrs, CT, USA. His general research interests lie in the areas of digital signal processing and

wireless communications, and his recent research has focused on underwater acoustic communications, sonar signal processing, and audio signal power amplifications.



Yi Huang (S'14) received the B.S. degree in electrical information engineering and the M.S. degree in signal and information processing from Northwestern Polytechnical University (NPU), Xi'an, China, in 2008 and 2011, respectively. He is currently working toward the Ph.D. degree in the Department of Electrical and Computer Engineering at the University of Connecticut (UConn), Storrs, CT, USA.

His research interests are in the areas of communications and signal processing. His recent focus has been on sparse channel estimation of underwater acoustic communications.



Shengli Zhou (S'99–M'03–SM'11–F'14) received the B.S. degree and M.Sc. degrees in electrical engineering and information science from the University of Science and Technology of China (USTC), Hefei, China, in 1995 and 1998, respectively, and the Ph.D. degree in electrical engineering from the University of Minnesota (UMN), Minneapolis, MN, USA, in 2002.

Currently, he is a full Professor with the Department of Electrical and Computer Engineering, University of Connecticut (UConn), Storrs, CT, USA.

He has held a United Technologies Corporation (UTC) Professorship in Engineering Innovation (2008–2011) and the Charles H. Knapp Associate Professor in Electrical Engineering (2012–2013). His general research interests lie in the areas of wireless communications and signal processing. His recent focus has been on underwater acoustic communications and networking.

Dr. Zhou has served as an Associate Editor for the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS (2005–2007) and the IEEE TRANSACTIONS ON SIGNAL PROCESSING (2008–2010), and he is currently an Associate Editor for the IEEE JOURNAL OF OCEANIC ENGINEERING. He received the 2007 U.S. Office of Naval Research (ONR) Young Investigator award and the 2007 Presidential Early Career Award for Scientists and Engineers (PECASE). He is a member of the Connecticut Academy of Science and Engineering (CASE).

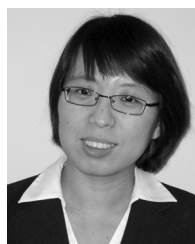


Zhijie Shi (M'04) received the B.S. and M.S. degrees in computer science from Tsinghua University, Beijing, China, in 1996 and 1992, respectively, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 2004.

Currently, he is an Associate Professor of Computer Science and Engineering at the University of Connecticut, Storrs, CT, USA. His current research interests include underwater sensor networks, and sensor network security, hardware mechanisms for secure and reliable computing, side channel attacks

and countermeasures, and embedded system designs.

Dr. Shi received the U.S. National Science Foundation CAREER award in 2006. He is a member of the Association for Computing Machinery (ACM).



Jun-Hong Cui (M'03) received the B.S. degree in computer science from Jilin University, Jilin, China in 1995, the M.S. degree in computer engineering from the Chinese Academy of Sciences, Beijing, China, in 1998, and the Ph.D. degree in computer science from the University of California Los Angeles (UCLA), Los Angeles, CA, USA, in 2003.

Currently, she is on the faculty of the Computer Science and Engineering Department, University of Connecticut, Storrs, CT, USA. Her research interests

cover the design, modeling, and performance evaluation of networks and distributed systems. Recently, her research has mainly focused on exploiting the spatial properties in the modeling of network topology, network mobility, and group membership; scalable and efficient communication support in overlay and peer-to-peer networks; and algorithm and protocol design in underwater sensor networks.

Prof. Cui is a member of the Association for Computing Machinery (ACM), ACM SIGCOMM, ACM SIGMOBILE, the IEEE Computer Society, and the IEEE Communications Society. She is actively involved in the community as an organizer, a Technical Program Committee (TPC) member, and a reviewer for many conferences and journals. She cofounded the ACM International Conference on UnderWater Networks and Systems (WUWNet), and is now serving as the WUWNet Steering Committee Chair.